



CS-Lab No. 6

UART and SPI Communication

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I. Objective

The goal of this prelab is to explore the fundamental aspects of communication protocols by exploring two well-known examples.

II. Concepts and Ideas

II.1. UART Communication

Universal Asynchronous Receiver-Transmitter (UART) is a widely used serial communication protocol. It enables two devices to communicate by exchanging data bytes asynchronously, without needing a shared clock signal. Key concepts include:

- Start and Stop Bits: Used to signal the beginning and end of data frames.
- Parity Bits: Optional error-checking mechanism to ensure data integrity.
- Baud Rate: Defines the speed of communication.

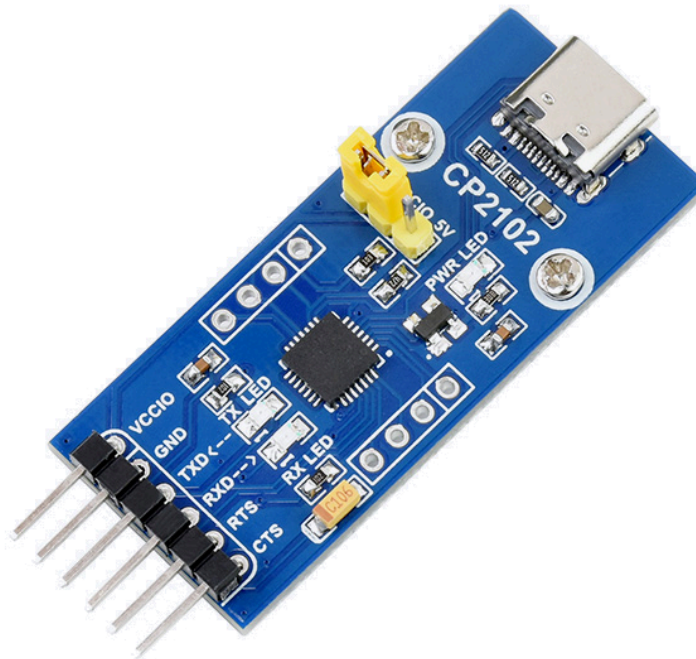


Fig. 1 — A USB to UART Converter

Usage Examples

Some common applications of UART include:

1. Debugging microcontroller applications via serial output.
2. Communication between GPS modules and a microcontroller.
3. Connecting Bluetooth or WiFi modules to embedded systems.

II.2. SPI Communication

Serial Peripheral Interface (SPI) is a synchronous protocol designed for fast data transfers between a master device and one or more slave devices. It uses the following signals:

- SCLK (Clock): Synchronizes data transfer.
- MOSI (Master Out Slave In): Sends data from master to slave.
- MISO (Master In Slave Out): Sends data from slave to master.
- SS/CS (Slave Select or Chip Select): Enables communication with specific slave devices.

SPI supports multiple devices on the same bus, using independent SS/CS signals.

Note

After the BLM movement in the US, this terminology for communication systems has become obsolete. You may see phrases such as «Tx or COPI» and «Rx and CIPO» instead of «MOSI» and «MISO» in newer designs (such as our raspberry pi pico). The original naming scheme was a tragedy, and the new ones makes the previous scheme seem like a good choice.

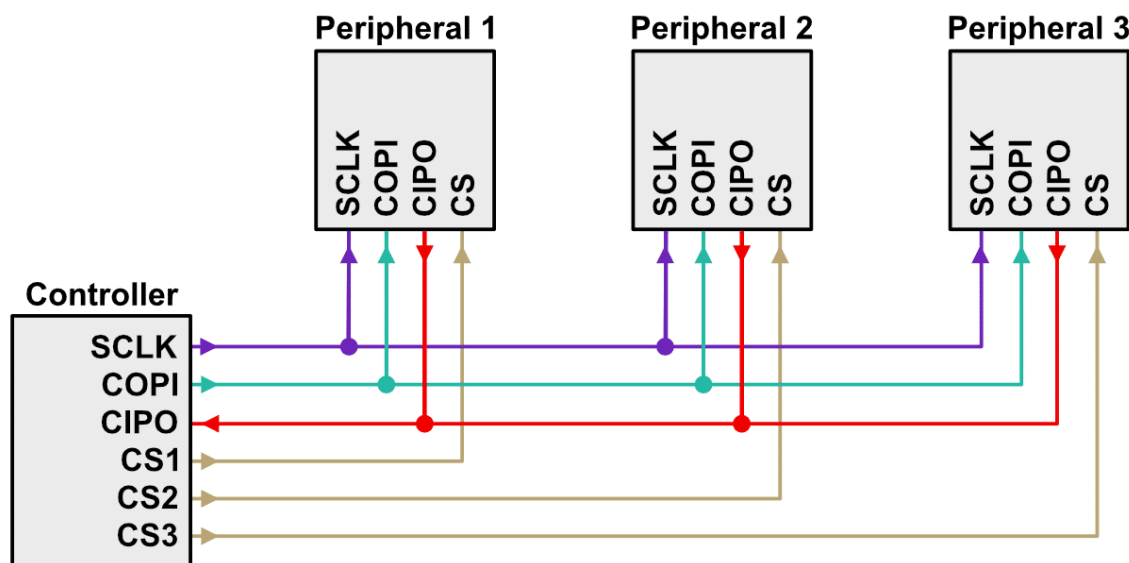


Fig. 2 — Multi Device SPI Communication Example. COPI=Controller Out Peripheral In

Clock Phase and Polarity

Clock phase (CPHA) and polarity (CPOL) determine how data is sampled and shifted during communication. CPOL indicates whether the clock is idle high or low, while CPHA specifies whether data is sampled on the leading or trailing edge of the clock pulse.

III. Tasks

1. UART Communication

- Draw the timing diagram of a UART communication sequence (without RTS and CTS signals). Label start bits, data bits, stop bits, and any parity bits.
- Decode the timing diagram you created and explain the meaning of the various components (e.g., stop bits and parity bits).

2. SPI Communication

- Name three sensors that utilize SPI and briefly explain their functionalities. You may look at the appendix-A part of the c-sdk reference document for this.
- Draw a timing diagram for SPI communication. You are free to select a CPOL and CPHA scheme of your liking.

- Label the roles of each SPI signal pin (SCLK, MOSI, MISO, SS/CS) on the timing diagram.

IV. Resources

Check [Sparkfun](#) . They have interesting write-ups on both UART and SPI (among other things). You can even burrow the timing diagrams from them!